FAMILY HABIT APP · DATASTORE · SPEC REVIEW · 1 NOV 2026, 16:00 SGT

Datastore spec review

This is what I will write into the specs once you approve it: **14 new rules**, **2 changed rules**, and the table rules as **plain-English tests**. It is based on draft 3 and your decisions A to M. Nothing is written yet.

1. Rules: new and changed
2. Other records
3. The table rules as tests

PART 1

Rules: new and changed

Each rule gets a permanent ID in its module's spec. A changed rule keeps its ID; the old wording is kept and marked SUPERSEDED. Every new rule starts as "Not yet tested" until the build.

ID	STATUS	SPEC FILE	RULE, IN PLAIN ENGLISH
1.4	changed	`docs/SPEC.md`	Every adjustable number lives **once, in code**, labelled with one of **five kinds: Tuneable, Named style, Named rule, Guardrail, Fixed** (replaces "guardrail, tunable, trial-plan"). The **Test options** for every kind also live in code (decision A). Trial-plan values such as the trial length live in code too, but are not experiment values and are not in the list of values (decision I). `src/experiments/EXPERIMENTS.md` lists the **31 values** in words (the 29 plus 30, Weekly Wrap Up on by default, and 31, Wrap Up day).
11.10	changed	`src/experiments/SPEC.md`	Values come in the **five kinds** of rule 1.4 (replaces "number, named style, named rule"). A live experiment has one **Control (Option A)**, the everyday value, and one or more **Test options (B, C…)**, all written in code through a reviewed pull request (`options_approved_to_try`). A Guardrail's Test options may only be stricter; a Fixed value has none.

ID	STATUS	SPEC FILE	RULE, IN PLAIN ENGLISH
8.3	new	`src/datastore/SPEC.md`	**The tables.** 14 are built now, in four groups: About each parent (`parents` , `children` , `consents` , `picks` , `slots`); What happened (`habit_logs` , `messages_sent` , `connection_answers` , `events`); App look up settings (`settings_versions` , `event_types` , `holiday_dates`); Housekeeping (`invites` , `deletion_log`). Six come later: four experiment tables before alpha, and album entries and badges with Variable rewards. Each table's columns are listed in the spec.
8.4	new	`src/datastore/SPEC.md`	**Add only, with three narrow edits.** Our server may only add rows, except: (1) change a parent's paused, Wrap Up on or off, and Wrap Up time; (2) fill in a pick's "stopped at" once; (3) replace a parent's slots. Each edit also writes an event. Only "delete my data" removes a parent's rows. Reference lists are read only. Enforced by database permissions, not just code.
8.5	new	`src/datastore/SPEC.md`	**Delete follows the parent.** Every personal table links to `parents` , so deleting a parent deletes all their rows. A test fails if a new table with a parent link isn't covered. Order: photos, then chat memory, then rows, then one deletion-log row (rule 1.5).
8.6	new	`src/datastore/SPEC.md`	**No duplicates.** A Telegram message is saved once only; one nudge message per parent, day and slot; one Wrap Up per parent per week; one connection answer per parent per week (the first counts); one current pick per parent per nudge; no duplicate slot; a settings version is never created twice for the same values.
8.7	new	`src/datastore/SPEC.md`	**All or nothing.** Each action that touches more than one table (for example "save a log and its event") runs as one database step: it all happens, or none of it does.
8.8	new	`src/datastore/SPEC.md`	**Structure only.** The database checks links, required fields, duplicates and Fixed limits (280 characters). Tuneable numbers such as "3 a day" are never repeated in the database (rule 1.4).
8.9	new	`src/datastore/SPEC.md`	**Links use the nudge's ID number** (v1.6). A nudge that any parent has picked can't be deleted, only retired (fulfils v1.5 answer 2).

ID	STATUS	SPEC FILE	RULE, IN PLAIN ENGLISH
8.10	new	`src/datastore/SPEC.md`	**Children.** Optional: one row per child with a **birth year** only (no name, month or school). A year in the future or more than 25 years ago is refused at sign-up and again in the saving code. Ages 0 to 6 and 13 to 25 are accepted and flagged (rule 10.4). The table is created now, empty in week one; parents are asked from alpha.
8.11	new	`src/datastore/SPEC.md`	**Least access for Telemetry.** Telemetry reads through views with no first names, Telegram IDs or descriptions.
8.12	new	`src/datastore/SPEC.md`	**Labelled and timed.** Every table has a "created at" time and a plain-English note on every column. Times are stored one way: a moment = time with time zone; a Singapore day = a date; a slot = minutes after midnight.
6.2	new	`src/photo-check/SPEC.md`	**A photo that fails the check is kept.** It is stored and shown to the parent in the Web app; the parent is asked to self-report, and the log is recorded as self-reported (rule 1.3).
5.6	new	`src/telegram-bot/SPEC.md`	**Weekly connection question.** After the first tap, the bot removes the 1 to 5 buttons and shows "Thanks, you answered 4". Any extra tap that still arrives is ignored (rule 8.6). A parent can't change the answer that week.
10.4	new	`src/telemetry/SPEC.md`	**Flag children outside 7 to 12.** Telemetry flags a child aged 0 to 6 or 13 to 25, rather than the app refusing them at sign-up.
11.13	new	`src/experiments/SPEC.md`	**Opt-out lives in consents.** Opting out of experiments is an add-only row in `consents` (topic: experiments), not in `assignments`. An opted-out parent gets every Control value, including in experiments started later.

PART 2

Other records

- `src/datastore/SPEC.md` : the "Planned for the Datastore build" notes are marked SUPERSEDED and point to rules 8.3 to 8.12, with each table's columns and its server permissions written out.
- `docs/SPEC.md` : a Version 1.12 change note listing every decision from today (the review, photos, missing tables, child birth year, A to M).
- `src/experiments/EXPERIMENTS.md` : the list of 31 values with their kinds (Tuneable, Named style, Named rule, Guardrail, Fixed).
- `docs/PRODUCT-BACKLOG.md` : the "what's running" timeline (F), owned by Experiments with a link from Telemetry, before alpha.
- `src/web-app/SPEC.md` : the two parked questions (how the Web app reaches the database; whether descriptions go to Claude, and what sign-up says), to decide when you design the Web app.
- **Decision L** (move the frequency choices and the 2-day floor into `src/settings.ts`) is done during the build, not written as a rule.
- `docs/LEARNING-LOG.md` : a short entry on Control and Test, settings versions, and "add-only" tables.

PART 3

The table rules as tests

Each line is one behaviour a test will check. They are written before any code (test-driven development). Most are **database tests**: they run against a local copy of the database with `npm run test:db` , which isn't yet part of the automatic checks (a known issue). Lines marked `[code]` are app tests that run in `npm run check` .

Access and permissions `rules 8.2, 8.4`

1. No table can be read or written by the public or by a signed-out visitor.
2. The server can add a parent, and can change only paused, Wrap Up on or off, and Wrap Up time. Changing their Telegram ID, name, sign-up date or cohort is refused.
3. The server can't edit or delete any row in `habit_logs` , `messages_sent` , `connection_answers` , `events` , `consents` , `children` or `settings_versions` .
4. The server can only read `event_types` and `holiday_dates` .
5. The server can't write to `deletion_log` directly; only "delete my data" can.

6. Telemetry's views show no first names, Telegram IDs or descriptions (rule 8.11).

Parents and children `rules 8.6, 8.10`

7. Two parents with the same Telegram ID are refused.
8. A parent with no children rows is normal: nothing is guessed.
9. A child needs a parent and a birth year; no name column exists.
10. `[code]` A birth year in the future, or more than 25 years ago, is refused when saving.
11. `[code]` A child aged 0 to 6 or 13 to 25 is saved, and flagged for Telemetry (rule 10.4).

Consents `rules 8.4, 11.13`

12. The topic must be recording, photos or experiments.
13. `[code]` The latest answer per topic counts; opting out of experiments means the parent gets every Control value.

Picks and slots `rules 8.4, 8.6, 8.9`

14. A pick must point at a real nudge; an unknown nudge is refused.
15. Only one current pick per parent per nudge.
16. "Stopped at" can't be before "chosen at", and once filled in it can't be changed.
17. A nudge that any parent has picked can't be deleted.
18. The same slot time twice on the same schedule is refused.
19. Replacing a parent's slots also writes an event.

Habit logs `rules 8.6, 8.7, 6.2, 5.4`

20. A description longer than 280 characters is refused.
21. A log for a day before yesterday is refused.
22. The same Telegram message saved twice creates only one log.
23. A log after a Done tap must point at a nudge message sent to the same parent.
24. A photo that failed the check is stored, and the log is recorded as self-reported.
25. If saving the log's event fails, the log isn't saved either (all or nothing).

Messages sent `rule 8.6`

26. A nudge message must name a nudge and an example; a Weekly Wrap Up must not.

27. A second nudge message for the same parent, day and slot is refused, even if two scheduler runs overlap.
28. A second Weekly Wrap Up for the same parent in the same week is refused.

Weekly connection answers `rules 8.6, 5.6`

29. The score must be 1 to 5.
30. A second answer from the same parent in the same week is ignored; the first counts.

Events and settings versions `rules 8.6, 11.1, 11.2`

31. An event must use a type from `event_types`; free text is refused.
32. An event's settings version and experiment stamp are never empty; in week one the stamp says "baseline".
33. The same Telegram message saved twice creates only one event.
34. Saving the same settings twice creates only one version; a version can't be edited.

Reference lists and invites `rules 1.1, 8.6`

35. The same holiday date twice is refused.
36. Invite codes are unique, and each one can be used only once.

"Delete my data" `rules 1.5, 8.5`

37. Deleting a parent removes every one of their rows in every personal table.
38. Other parents' rows are untouched.
39. One deletion-log row is written, with the day and a count, and no name or ID.
40. An invite the parent used stays marked as used, but no longer points to them.
41. A test fails if any table with a parent link isn't covered by the deletion.

Not tested in this build: the bot removing the buttons (rule 5.6, tested when the Telegram bot is built), Telemetry's flags (rule 10.4, tested with Telemetry), photo storage itself (with the Photo check), and the experiment tables (before alpha).

Prepared by Claude, 1 Nov 2026 16:00 SGT, for the owner of the Family Habit App. Based on Datastore tables, draft 3 and your decisions A to M. Nothing has been written to the project yet.